

MYSQL Testing

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Case Study- JDBC Connection through MySQL Server

Using 3 interfaces , connections , statements and results , execute selenium automation database using

SELECT *

Steps -

Step 1 - Establish the connection with the SQL Database

```
Database changed
mysql> show tables;
+-----+
| Tables_in_selenium_automation |
+-----+
| book_detl                      |
| CUST_NEW                      |
| ORDER_NEW                     |
| stu_detl                      |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> select * from book_detl;
+-----+-----+-----+-----+-----+-----+-----+-----+
| book_no | title    | sub_code | author  | publisher | status | yop  | price |
+-----+-----+-----+-----+-----+-----+-----+-----+
| BI001   | Clones   | Bio      | Ian     | XYZ       | S      | 1997 | 700   |
| BI002   | DNA      | Bio      | Watson  | XYZ       | S      | unkn | 575   |
| CH001   | Elements | Che      | Robert  | XYZ       | S      | 1890 | 250   |
| CH002   | Nucleus  | Che      | Hilburn | XYZ       | S      | 1992 | 350   |
| MA001   | Calculus | Mat      | Newton  | PQR       | S      | UNKN | 195   |
| MA002   | Algebra  | Mat      | Raman   | PQR       | S      | 1990 | 190   |
| MA003   | Vectors  | Mat      | Richard | PQR       | S      | unkn | 200   |
| PH001   | Chaos    | Phy      | Feynman | ABC       | S      | 1997 | 300   |
| PH002   | Relativity | Phy     | Einstein | ABC       | S      | 1902 | 400   |
+-----+-----+-----+-----+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

Step 1.1 - Add Dependencies in maven project pom.xml

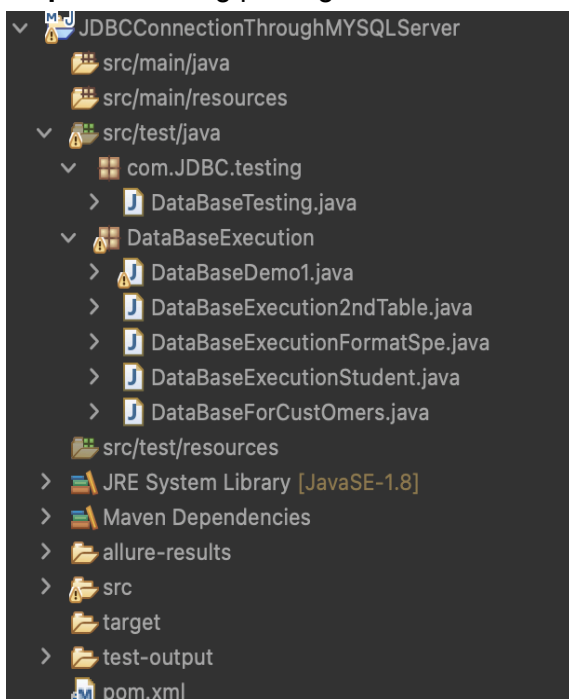
```
JDBCConnectionThroughMySQLServer/pom.xml X DataBaseForCustOmers.java DataBaseTesting.java DataBaseDemo1.java
2  xmlns:xsi="https://www.w3.org/2001/XMLSchema-instance"
3  xsi:schemaLocation="https://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4  <modelVersion>4.0.0</modelVersion>
5  <groupId>JDBCConnectionThrouhMySQLServer</groupId>
6  <artifactId>JDBCConnectionThrouhMySQLServer</artifactId>
7  <version>0.0.1-SNAPSHOT</version>
8  <name>Sushank</name>
9  <dependencies>
10 <!-- Source:
11 https://mvnrepository.com/artifact/com.mysql/mysql-connector-j -->
12 <dependency>
13 <groupId>com.mysql</groupId>
14 <artifactId>mysql-connector-j</artifactId>
15 <version>26.7.0</version>
16 <scope>compile</scope>
17 </dependency>
18
19 <!-- Source: https://mvnrepository.com/artifact/org.json/json -->
20 <dependency>
21 <groupId>org.json</groupId>
22 <artifactId>json</artifactId>
23 <version>20260522</version>
24 <scope>compile</scope>
25 </dependency>
```

```
26  <!--
27  Source:https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
28  <dependency>
29    <groupId>org.seleniumhq.selenium</groupId>
30    <artifactId>selenium-java</artifactId>
31    <version>4.45.0</version>
32    <scope>compile</scope>
33  </dependency>
34
35  <dependency>
36    <groupId>org.apache.poi</groupId>
37    <artifactId>poi</artifactId>
38    <version>4.1.2</version>
39  </dependency>
40
41  <dependency>
42    <groupId>org.apache.poi</groupId>
43    <artifactId>poi-ooxml</artifactId>
44    <version>4.1.2</version>
45  </dependency>
46  <!-- Source: https://mvnrepository.com/artifact/commons-io/commons-io -->
47  <dependency>
48    <groupId>commons-io</groupId>
49    <artifactId>commons-io</artifactId>
50    <version>2.22.0</version>
51    <scope>compile</scope>
52  </dependency>
53
54  <!-- Source: https://mvnrepository.com/artifact/net.sf.opencsv/opencsv -->
55  <dependency>
56    <groupId>net.sf.opencsv</groupId>
57    <artifactId>opencsv</artifactId>
58    <version>2.3</version>
59    <scope>compile</scope>
60  </dependency>
61  <!-- Source:
62  https://mvnrepository.com/artifact/io.qameta.allure/allure-testing -->
63  <dependency>
64    <groupId>io.qameta.allure</groupId>
65    <artifactId>allure-testing</artifactId>
66    <version>2.35.1</version>
67    <scope>compile</scope>
68  </dependency>
69  <!-- Source: https://mvnrepository.com/artifact/org.testng/testng -->
70  <dependency>
71    <groupId>org.testng</groupId>
72    <artifactId>testng</artifactId>
73    <version>7.12.0</version>
74    <scope>test</scope>
```

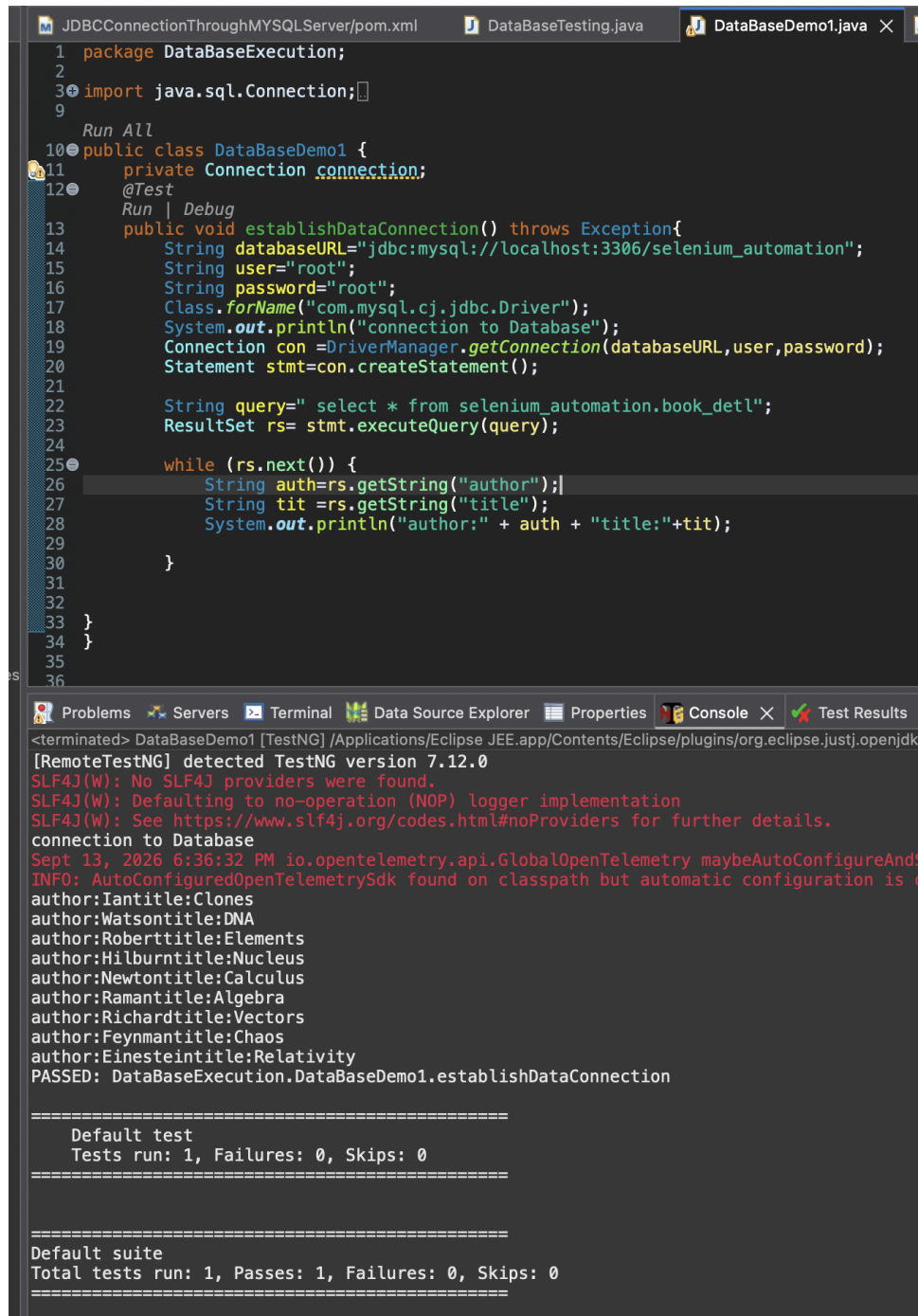
Overview Dependencies Dependency Hierarchy Effective POM pom.xml

Problems Servers Terminal Data Source Explorer Properties Console Test Results JUnit Results of running ci

Step 1.2 - Adding packages and classes



Step 1.3 - Execution Set for Results Query in Eclipse



```
1 package DataBaseExecution;
2
3 import java.sql.Connection;
4
5
6
7
8
9
10 public class DataBaseDemo1 {
11     private Connection connection;
12     @Test
13     public void establishDataConnection() throws Exception{
14         String databaseURL="jdbc:mysql://localhost:3306/selenium_automation";
15         String user="root";
16         String password="root";
17         Class.forName("com.mysql.cj.jdbc.Driver");
18         System.out.println("connection to Database");
19         Connection con =DriverManager.getConnection(databaseURL,user,password);
20         Statement stmt=con.createStatement();
21
22         String query=" select * from selenium_automation.book_detl";
23         ResultSet rs= stmt.executeQuery(query);
24
25         while (rs.next()) {
26             String auth=rs.getString("author");
27             String tit =rs.getString("title");
28             System.out.println("author:" + auth + "title:"+tit);
29         }
30     }
31 }
32
33 }
34 }
35
36
```

Problems Servers Terminal Data Source Explorer Properties Console Test Results

```
<terminated> DataBaseDemo1 [TestNG] /Applications/Eclipse JEE.app/Contents/Eclipse/plugins/org.eclipse.justj.openjdk
[RemoteTestNG] detected TestNG version 7.12.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
connection to Database
Sept 13, 2026 6:36:32 PM io.opentelemetry.api.GlobalOpenTelemetry maybeAutoConfigureAndS
INFO: AutoConfiguredOpenTelemetrySdk found on classpath but automatic configuration is d
author:Ian title:Clones
author:Watson title:DNA
author:Robert title:Elements
author:Hilburn title:Nucleus
author:Newton title:Calculus
author:Ramant title:Algebra
author:Richard title:Vectors
author:Feynman title:Chaos
author:Einstein title:Relativity
PASSED: DataBaseExecution.DataBaseDemo1.establishDataConnection

=====
Default test
Tests run: 1, Failures: 0, Skips: 0
=====

=====
Default suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
=====
```

Step 1.4 - Execution in Specific Format

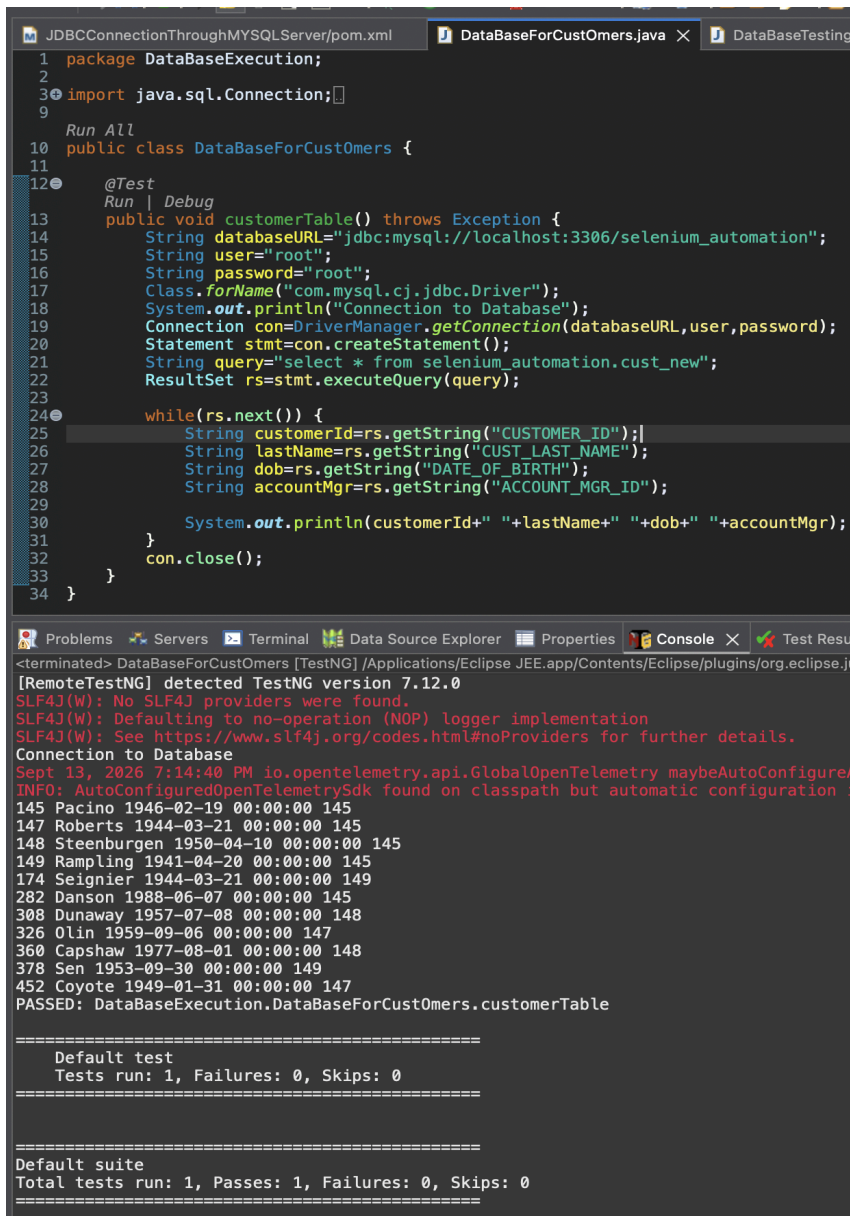
Step 2 - Connection of SQL database in MYSQL Server and Execution of 2nd Table cust_new

```
mysql> select * from cust_new;
```

CUSTOMER_ID	CUST_LAST_NAME	DATE_OF_BIRTH	ACCOUNT_MGR_ID
145	Pacino	1946-02-19 00:00:00	145
147	Roberts	1944-03-21 00:00:00	145
148	Steenburgen	1950-04-10 00:00:00	145
149	Rampling	1941-04-20 00:00:00	145
174	Seignier	1944-03-21 00:00:00	149
282	Danson	1988-06-07 00:00:00	145
308	Dunaway	1957-07-08 00:00:00	148
326	Olin	1959-09-06 00:00:00	147
360	Capshaw	1977-08-01 00:00:00	148
378	Sen	1953-09-30 00:00:00	149
452	Coyote	1949-01-31 00:00:00	147

```
11 rows in set (0.00 sec)
```

Step 2.1 - Execution Set for Results Query in Eclipse



```
1 package DataBaseExecution;
2
3 import java.sql.Connection;
4
5
6
7
8
9
10 public class DataBaseForCustOmers {
11
12     @Test
13     public void customerTable() throws Exception {
14         String databaseURL="jdbc:mysql://localhost:3306/selenium_automation";
15         String user="root";
16         String password="root";
17         Class.forName("com.mysql.cj.jdbc.Driver");
18         System.out.println("Connection to Database");
19         Connection con=DriverManager.getConnection(databaseURL,user,password);
20         Statement stmt=con.createStatement();
21         String query="select * from selenium_automation.cust_new";
22         ResultSet rs=stmt.executeQuery(query);
23
24         while(rs.next()) {
25             String customerId=rs.getString("CUSTOMER_ID");
26             String lastName=rs.getString("CUST_LAST_NAME");
27             String dob=rs.getString("DATE_OF_BIRTH");
28             String accountMgr=rs.getString("ACCOUNT_MGR_ID");
29
30             System.out.println(customerId+" "+lastName+" "+dob+" "+accountMgr);
31         }
32         con.close();
33     }
34 }
```

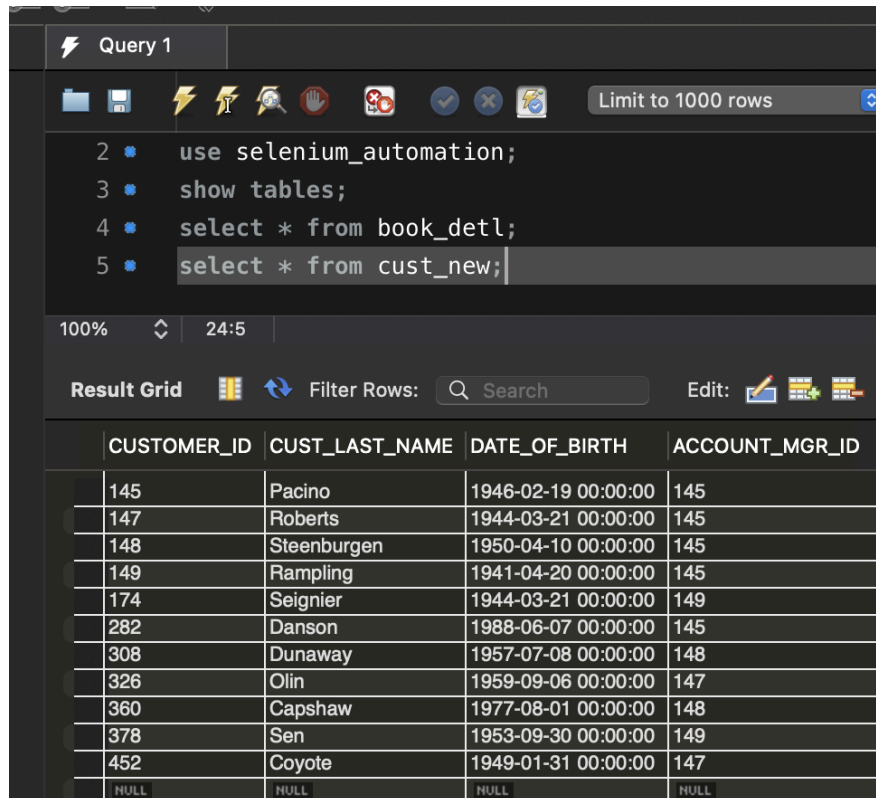
Problems Servers Terminal Data Source Explorer Properties Console Test Results

```
<terminated> DataBaseForCustOmers [TestNG] /Applications/Eclipse JEE.app/Contents/Eclipse/plugins/org.eclipse.jdt
[RemoteTestNG] detected TestNG version 7.12.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
Connection to Database
Sept 13, 2026 7:14:40 PM io.opentelemetry.api.GlobalOpenTelemetry maybeAutoConfigure
INFO: AutoConfiguredOpenTelemetrySdk found on classpath but automatic configuration
145 Pacino 1946-02-19 00:00:00 145
147 Roberts 1944-03-21 00:00:00 145
148 Steenburgen 1950-04-10 00:00:00 145
149 Rampling 1941-04-20 00:00:00 145
174 Seignier 1944-03-21 00:00:00 149
282 Danson 1988-06-07 00:00:00 145
308 Dunaway 1957-07-08 00:00:00 148
326 Olin 1959-09-06 00:00:00 147
360 Capshaw 1977-08-01 00:00:00 148
378 Sen 1953-09-30 00:00:00 149
452 Coyote 1949-01-31 00:00:00 147
PASSED: DataBaseExecution.DataBaseForCustOmers.customerTable

=====
Default test
Tests run: 1, Failures: 0, Skips: 0
=====

Default suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
=====
```

Step 2.2 - Connection in MySQL- WorkBench and Execution



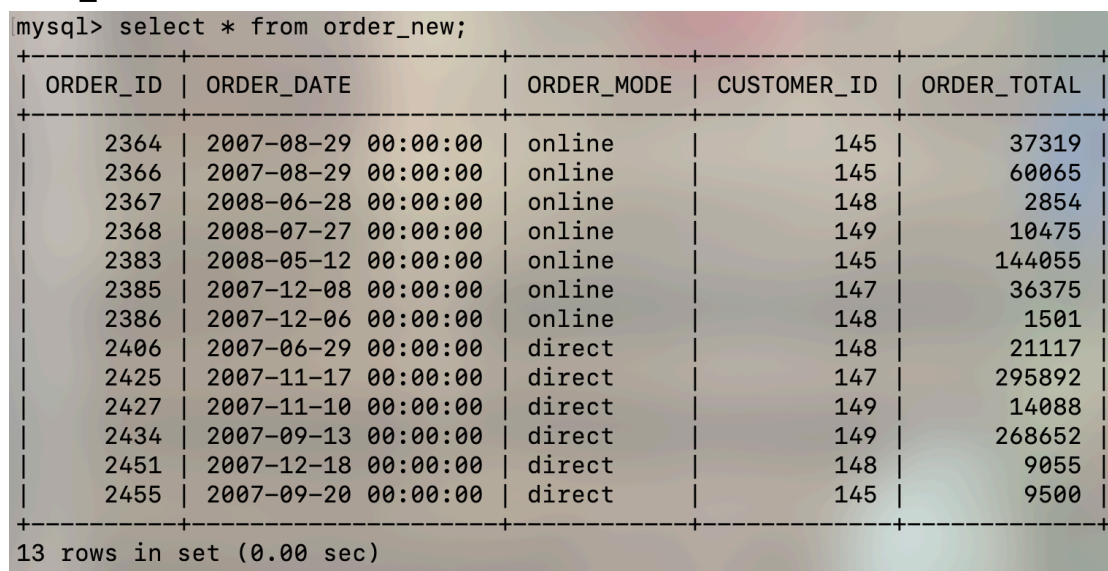
The screenshot shows the MySQL Workbench interface. The Query Editor at the top contains the following SQL statements:

```
2 use selenium_automation;
3 show tables;
4 select * from book_detl;
5 select * from cust_new;
```

Below the editor, the Result Grid displays the output of the last query. It shows a table with 4 columns: CUSTOMER_ID, CUST_LAST_NAME, DATE_OF_BIRTH, and ACCOUNT_MGR_ID. The data is as follows:

CUSTOMER_ID	CUST_LAST_NAME	DATE_OF_BIRTH	ACCOUNT_MGR_ID
145	Pacino	1946-02-19 00:00:00	145
147	Roberts	1944-03-21 00:00:00	145
148	Steenburgen	1950-04-10 00:00:00	145
149	Rampling	1941-04-20 00:00:00	145
174	Seignier	1944-03-21 00:00:00	149
282	Danson	1988-06-07 00:00:00	145
308	Dunaway	1957-07-08 00:00:00	148
326	Olin	1959-09-06 00:00:00	147
360	Capshaw	1977-08-01 00:00:00	148
378	Sen	1953-09-30 00:00:00	149
452	Coyote	1949-01-31 00:00:00	147
NULL	NULL	NULL	NULL

Step 3 - connection of SQL database in MYSQL Server and Execution of 3rd Table Order_new



The screenshot shows a MySQL command line session. The query executed is:

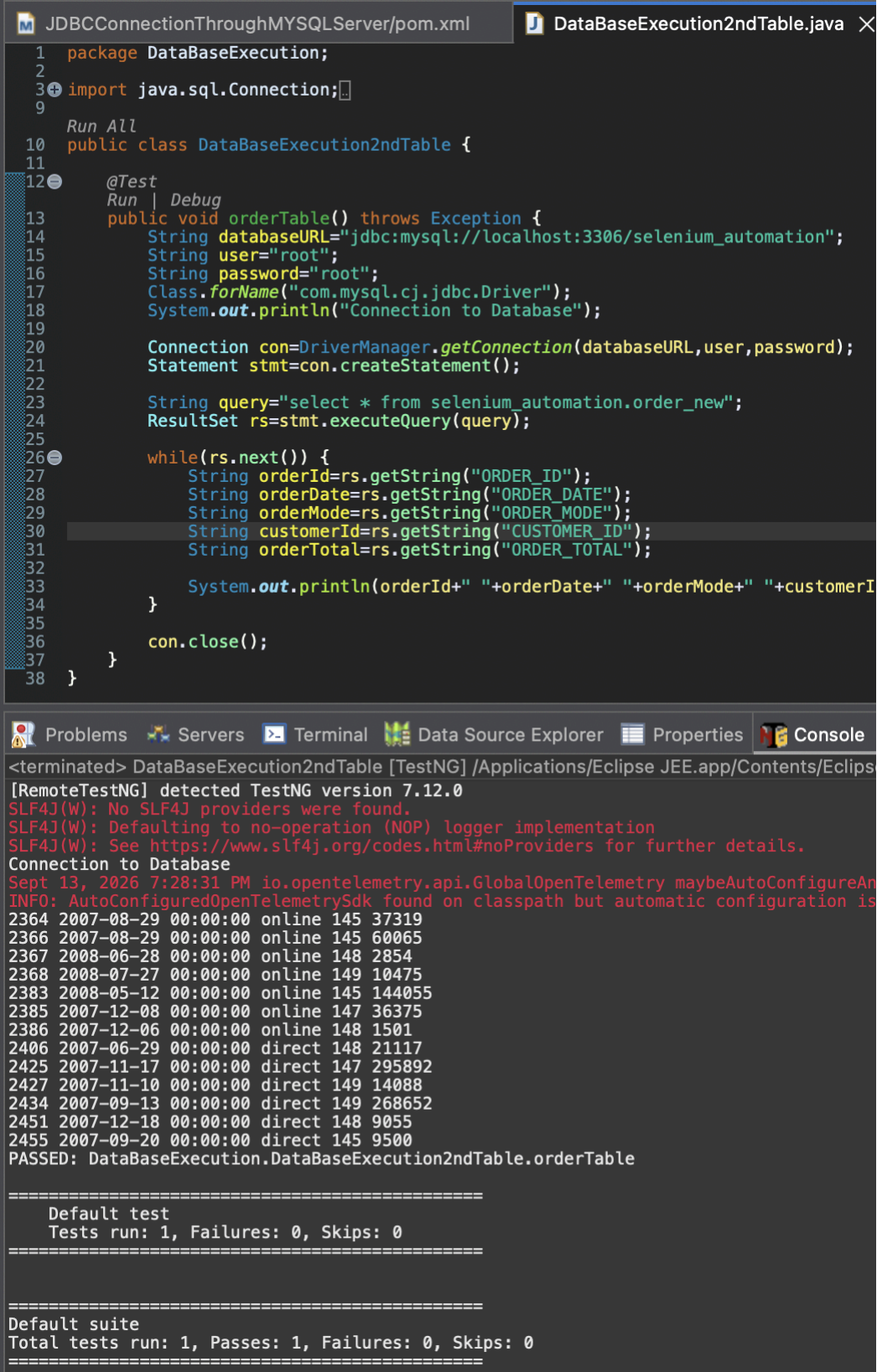
```
mysql> select * from order_new;
```

The output is a table with 5 columns: ORDER_ID, ORDER_DATE, ORDER_MODE, CUSTOMER_ID, and ORDER_TOTAL. The data is as follows:

ORDER_ID	ORDER_DATE	ORDER_MODE	CUSTOMER_ID	ORDER_TOTAL
2364	2007-08-29 00:00:00	online	145	37319
2366	2007-08-29 00:00:00	online	145	60065
2367	2008-06-28 00:00:00	online	148	2854
2368	2008-07-27 00:00:00	online	149	10475
2383	2008-05-12 00:00:00	online	145	144055
2385	2007-12-08 00:00:00	online	147	36375
2386	2007-12-06 00:00:00	online	148	1501
2406	2007-06-29 00:00:00	direct	148	21117
2425	2007-11-17 00:00:00	direct	147	295892
2427	2007-11-10 00:00:00	direct	149	14088
2434	2007-09-13 00:00:00	direct	149	268652
2451	2007-12-18 00:00:00	direct	148	9055
2455	2007-09-20 00:00:00	direct	145	9500

13 rows in set (0.00 sec)

Step 3.1 - Execution Set for Results Query in Eclipse



The screenshot displays the Eclipse IDE interface. The top editor shows the file `DataBaseExecution2ndTable.java` with the following code:

```
1 package DataBaseExecution;
2
3 import java.sql.Connection;
4
5
6
7
8
9
10 public class DataBaseExecution2ndTable {
11
12     @Test
13     public void orderTable() throws Exception {
14         String databaseURL="jdbc:mysql://localhost:3306/selenium_automation";
15         String user="root";
16         String password="root";
17         Class.forName("com.mysql.cj.jdbc.Driver");
18         System.out.println("Connection to Database");
19
20         Connection con=DriverManager.getConnection(databaseURL,user,password);
21         Statement stmt=con.createStatement();
22
23         String query="select * from selenium_automation.order_new";
24         ResultSet rs=stmt.executeQuery(query);
25
26         while(rs.next()) {
27             String orderId=rs.getString("ORDER_ID");
28             String orderDate=rs.getString("ORDER_DATE");
29             String orderMode=rs.getString("ORDER_MODE");
30             String customerId=rs.getString("CUSTOMER_ID");
31             String orderTotal=rs.getString("ORDER_TOTAL");
32
33             System.out.println(orderId+" "+orderDate+" "+orderMode+" "+customerId+" "+orderTotal);
34         }
35
36         con.close();
37     }
38 }
```

The bottom console shows the output of the test execution:

```
<terminated> DataBaseExecution2ndTable [TestNG] /Applications/Eclipse JEE.app/Contents/Eclipse
[RemoteTestNG] detected TestNG version 7.12.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
Connection to Database
Sept 13, 2026 7:28:31 PM io.opentelemetry.api.GlobalOpenTelemetry maybeAutoConfigureAn
INFO: AutoConfiguredOpenTelemetrySdk found on classpath but automatic configuration is
2364 2007-08-29 00:00:00 online 145 37319
2366 2007-08-29 00:00:00 online 145 60065
2367 2008-06-28 00:00:00 online 148 2854
2368 2008-07-27 00:00:00 online 149 10475
2383 2008-05-12 00:00:00 online 145 144055
2385 2007-12-08 00:00:00 online 147 36375
2386 2007-12-06 00:00:00 online 148 1501
2406 2007-06-29 00:00:00 direct 148 21117
2425 2007-11-17 00:00:00 direct 147 295892
2427 2007-11-10 00:00:00 direct 149 14088
2434 2007-09-13 00:00:00 direct 149 268652
2451 2007-12-18 00:00:00 direct 148 9055
2455 2007-09-20 00:00:00 direct 145 9500
PASSED: DataBaseExecution.DataBaseExecution2ndTable.orderTable

=====
Default test
Tests run: 1, Failures: 0, Skips: 0
=====

=====
Default suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
=====
```

Step 3.2 - Connection in MySQL- WorkBench and Execution

The screenshot shows the MySQL Workbench interface. The top toolbar includes icons for file operations, execution, and settings. The 'Query 1' window contains the following SQL queries:

```
3 • show tables;
4 • select * from book_detl;
5 • select * from cust_new;
6 • select * from order_new;
```

The 'Result Grid' tab is active, displaying the results of the last query. The grid shows 16 rows of data with the following columns: ORDER_ID, ORDER_DATE, ORDER_MODE, CUSTOMER_ID, and ORDER_TOTAL.

ORDER_ID	ORDER_DATE	ORDER_MODE	CUSTOMER_ID	ORDER_TOTAL
2364	2007-08-29 00:00:00	online	145	37319
2366	2007-08-29 00:00:00	online	145	60065
2367	2008-06-28 00:00:00	online	148	2854
2368	2008-07-27 00:00:00	online	149	10475
2383	2008-05-12 00:00:00	online	145	144055
2385	2007-12-08 00:00:00	online	147	36375
2386	2007-12-06 00:00:00	online	148	1501
2406	2007-06-29 00:00:00	direct	148	21117
2425	2007-11-17 00:00:00	direct	147	295892
2427	2007-11-10 00:00:00	direct	149	14088
2434	2007-09-13 00:00:00	direct	149	268652
2451	2007-12-18 00:00:00	direct	148	9055
2455	2007-09-20 00:00:00	direct	145	9500
NULL	NULL	NULL	NULL	NULL

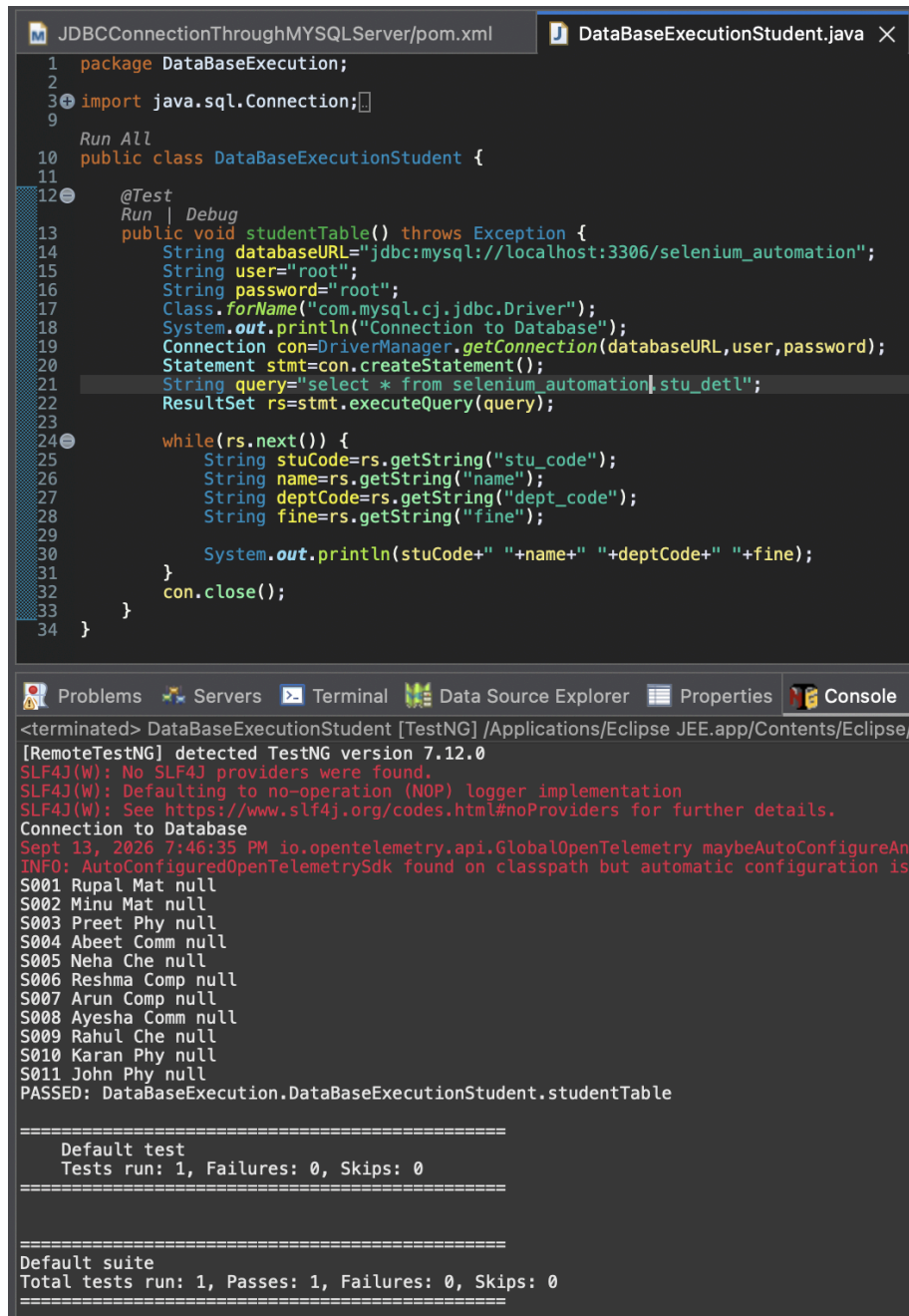
Step 4 - Connection of SQL database in MYSQL Server and Execution of 4th Table Stud_detl

```
mysql> select * from stu_detl;
```

stu_code	name	dept_code	fine
S001	Rupal	Mat	NULL
S002	Minu	Mat	NULL
S003	Preet	Phy	NULL
S004	Abeet	Comm	NULL
S005	Neha	Che	NULL
S006	Reshma	Comp	NULL
S007	Arun	Comp	NULL
S008	Ayesha	Comm	NULL
S009	Rahul	Che	NULL
S010	Karan	Phy	NULL
S011	John	Phy	NULL

11 rows in set (0.00 sec)

Step 4.2 - Execution Set for Results Query in Eclipse



The screenshot shows the Eclipse IDE with a Java project named 'JDBCConnectionThroughMySQLServer'. The file 'DataBaseExecutionStudent.java' is open, showing a Java class with a method 'studentTable()' that connects to a MySQL database and executes a query to retrieve student data. The console output shows the execution results, including a list of student records and test status messages.

```
1 package DataBaseExecution;
2
3 import java.sql.Connection;
4
5
6
7
8
9
10 Run All
11 public class DataBaseExecutionStudent {
12
13     @Test
14     public void studentTable() throws Exception {
15         String databaseURL="jdbc:mysql://localhost:3306/selenium_automation";
16         String user="root";
17         String password="root";
18         Class.forName("com.mysql.cj.jdbc.Driver");
19         System.out.println("Connection to Database");
20         Connection con=DriverManager.getConnection(databaseURL,user,password);
21         Statement stmt=con.createStatement();
22         String query="select * from selenium_automation.stu_detl";
23         ResultSet rs=stmt.executeQuery(query);
24
25         while(rs.next()) {
26             String stuCode=rs.getString("stu_code");
27             String name=rs.getString("name");
28             String deptCode=rs.getString("dept_code");
29             String fine=rs.getString("fine");
30
31             System.out.println(stuCode+" "+name+" "+deptCode+" "+fine);
32         }
33         con.close();
34     }
35 }
```

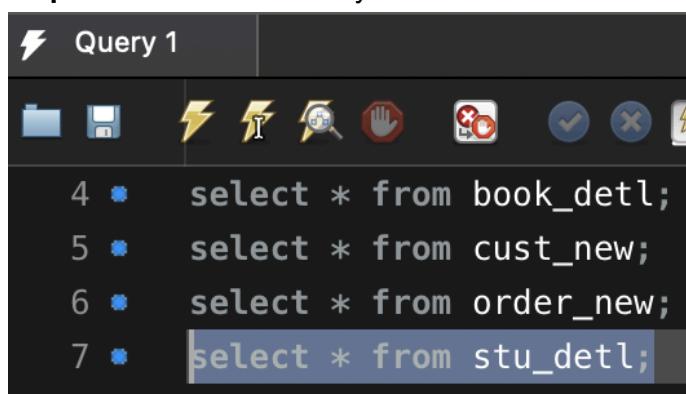
Console Output:

```
<terminated> DataBaseExecutionStudent [TestNG] /Applications/Eclipse JEE.app/Contents/Eclipse/
[RemoteTestNG] detected TestNG version 7.12.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
Connection to Database
Sept 13, 2026 7:46:35 PM io.opentelemetry.api.GlobalOpenTelemetry maybeAutoConfigureAn
INFO: AutoConfiguredOpenTelemetrySdk found on classpath but automatic configuration is
S001 Rupal Mat null
S002 Minu Mat null
S003 Preet Phy null
S004 Abeet Comm null
S005 Neha Che null
S006 Reshma Comp null
S007 Arun Comp null
S008 Ayesha Comm null
S009 Rahul Che null
S010 Karan Phy null
S011 John Phy null
PASSED: DataBaseExecution.DataBaseExecutionStudent.studentTable

=====
Default test
Tests run: 1, Failures: 0, Skips: 0
=====

Default suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
=====
```

Step 4.3 - Connection in MySQL- WorkBench and Execution



The screenshot shows the MySQL Workbench interface with a query editor. The editor contains four SQL queries, with the fourth query, 'select * from stu_detl;', highlighted.

```
4 select * from book_detl;
5 select * from cust_new;
6 select * from order_new;
7 select * from stu_detl;
```

100%  1:7

Result Grid   Filter Rows:

	stu_code	name	dept_code	fine	
	S001	Rupal	Mat	NULL	
	S002	Minu	Mat	NULL	
	S003	Preet	Phy	NULL	
	S004	Abeet	Comm	NULL	
	S005	Neha	Che	NULL	
	S006	Reshma	Comp	NULL	
	S007	Arun	Comp	NULL	
	S008	Ayesha	Comm	NULL	
	S009	Rahul	Che	NULL	
	S010	Karan	Phy	NULL	
	S011	John	Phy	NULL	
	NULL	NULL	NULL	NULL	